

Problem 45. Let ABC be a right-angled triangle with $\angle ACB = 90^\circ$. Denote by

- D the foot of the perpendicular from C to AB ;
- E and F the reflections of D about AC and BC , respectively;
- O_1 and O_2 the circumcentres of $\triangle ECB$ and $\triangle FCA$, respectively.

Prove that $AB = 2 \cdot O_1O_2$.

Problem 46. In an acute-angled triangle ABC , let D be the midpoint of BC and let P be a point on the segment AD . The interior angle bisectors of $\angle ABP$ and $\angle ACP$ intersect at Q . If $BQ \perp QC$, prove that Q lies on the segment AP .

Problem 47. Suppose 100 points in the plane are coloured using two colours, red and white, such that each red point is the centre of a circle passing through at least three white points. What is the least possible number of white points?

Problem 48. Find all ordered pairs (m, n) of integers so that the following quartic equation in x given by

$$x^4 + 2mx^2 + nx + m^2 - 36 = 0$$

has four integer roots. The roots are counted with multiplicity, i.e. double roots are counted twice, triple roots are counted thrice, etc.